

Experiment 6

Parallel Interfacing: Interfacing Seven Segment Display

CLOs Covered

CLO2: Perform hardware interfacing on a microcontroller platform effectively. (**PLO5/GA5**)

CLO5: Show professional behaviour while in the lab (**PLO8/GA8**)

Objective

This lab provides an opportunity to learn about the use of a microcontroller (MCU) and its interfacing to external devices. The lab uses the Tiva-C TM4C123 LaunchPad that has an ARM Cortex-M4F based TM4C123GH6PM MCU in it. This lab uses C language programming to drive a 7-segment display.

7-Segment Display Construction

A 7-Segment display is a useful electronic component use to produce numeric, alphabetic and some non-alphabetic symbols using a specific arrangement of LEDs as shown in Figure 6.1.

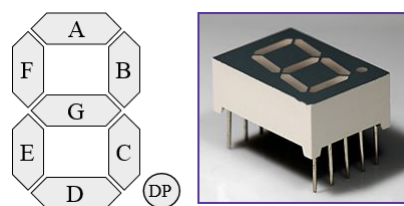


Figure 6.1: 7 Segment LED Display

A seven segment display consists of seven LEDs arranged in the form of a squarish '8' slightly inclined to the right and a single LED as the dot character. Different characters can be displayed by selectively glowing the required LED segments. Seven segment displays are of two types, common cathode and common anode. In common cathode type, the cathode of all LEDs are tied together to a single terminal which is usually labeled as 'com' and the anode of all LEDs are left alone as individual pins labeled as a, b, c, d, e, f, g & dot. In common anode type, the

anode of all LEDs are tied together as a single terminal and cathodes are left alone as individual pins. Both the configurations are shown in Figure 6.2

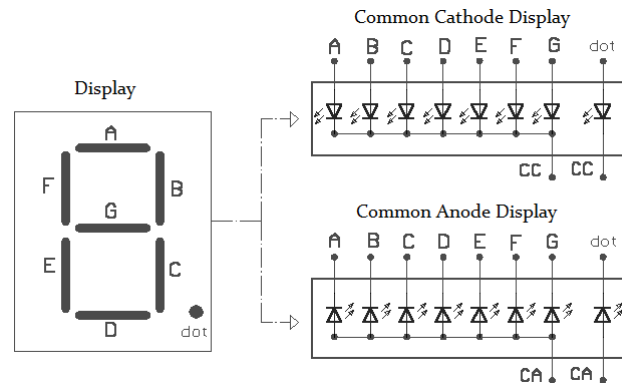


Figure 6.2: Common Anode and Common Cathode Configurations

In this experiment, a GPIO port on the TM4C123GH6PM MCU transmits 8 bits of data. These bits are applied to the seven-segment display to cause it to illuminate the appropriate segments to display the proper number or character. The seven segment used in this experiment is of common-anode type. Segment intensity is dependent on the current flow and should not exceed the limit of the segment.

Digit Drive Pattern

Digit drive pattern of a seven segment LED display is simply the different logic combinations of its terminals. For a certain character, a combination of LED ON and LED OFF is generated to display the character for a short period of time. The pattern is loaded alternately to display other characters. For example, to display the number 2, LEDs a, b, d, e, and g are illuminated. Table 6.3 provides the display pattern for numbers(0-9) and characters A through F.

Multiplexing the Seven Segments

To control four 7-segment displays, multiplexing can reduce the number of GPIO pins required. In this setup, the four multiplexed seven-segment displays are turned on one at a time to output the appropriate display. Because of the visual phenomenon known as *persistence of vision*, rapid switching of the seven-segment display can appear as if all four displays are turned on.

Launchpad Interface

Before you build your circuit, you should carefully note the non-sequential pin-out diagram of the LaunchPad shown in Figure 6.4. Although the TM4C123GH6PM MCU has 43 GPIO, only 35 of them are available through the LaunchPad. They are: 6 pins of Port A (PA2-PA7), 8 pins of Port B (PB0-PB7), 4 pins of Port C (PC4-PC7), 6 pins of Port D (PD0-PD3, PD6-PD7), 6 pins of Port E (PE0-PE5) and 5 pins of Port F (PF0-PF4). In addition, there are two ground, one 3.3V, one 5V (VBUS), and one reset pins available on the LaunchPad.

Characters	DP	G	F	E	D	C	B	A	Hexadecimal
0	1	1	0	0	0	0	0	0	0xC0
1	1	1	1	1	1	0	0	1	0xF9
2	1	0	1	0	0	1	0	0	0xA4
3	1	0	1	1	0	0	0	0	0xB0
4	1	0	0	1	1	0	0	1	0x99
5	1	0	0	1	0	0	1	0	0x92
6	1	0	0	0	0	0	1	0	0x82
7	1	1	1	1	1	0	0	0	0xF8
8	1	0	0	0	0	0	0	0	0x80
9	1	0	0	1	0	0	0	0	0x90
A	1	0	0	0	1	0	0	0	0x88
B	1	0	0	0	0	0	1	1	0x83
C	1	1	0	0	0	1	1	0	0xC6
D	1	0	1	0	0	0	0	1	0xA1
E	1	0	0	0	0	1	1	0	0x86
F	1	0	0	0	1	1	1	0	0x8E

Figure 6.3: 7 Segment Decoder Table

Pins PC0-PC3 are left off as they are used for JTAG debugging. Pins PA0-PA1 are also left off as they are used to create a virtual COM port to connect the LaunchPad to PC. These pins should not be used for regular I/O purpose.

	J1	J3		J4	J2		
3.3V	●	●	VBUS	PF2	●	●	GND
PB5	●	●	GND	PF3	●	●	PB2
PB0	●	●	PD0	PB3	●	●	PE0
PB1	●	●	PD1	PC4	●	●	PF0
PE4	●	●	PD2	PC5	●	●	RST
PE5	●	●	PD3	PC6	●	●	PB7
PB4	●	●	PE1	PC7	●	●	PB6
PA5	●	●	PE2	PD6	●	●	PA4
PA6	●	●	PE3	PD7	●	●	PA3
PA7	●	●	PF1	PF4	●	●	PA2

Figure 6.4: Header Pins on the Launchpad (EK-TM4C123GH6PM)

Display System Using Seven Segments

In this experiment, we will program the LaunchPad and four-digit 7-segment display module to display “UOFS” or “H”, “E”, “L”, “O” depending upon the function selected.

Note carefully the diagram of the display module. Connect the circuit as shown in Figure 6.5. The LaunchPad board is sufficient to supply +5V to Vdd of the 7-segment display module - connect the display power pins directly to VBUS and GND of J3. The experiment uses C language to program the MCU. Write a program to accomplish the task. You will need to open

a new project on Keil μ Vision. Note that display latches are connected to PA2-PA5 and the data pins are connected to PB0-PB7. Using the Latch Enable (LE) pin, your program can update the display digits one at a time.

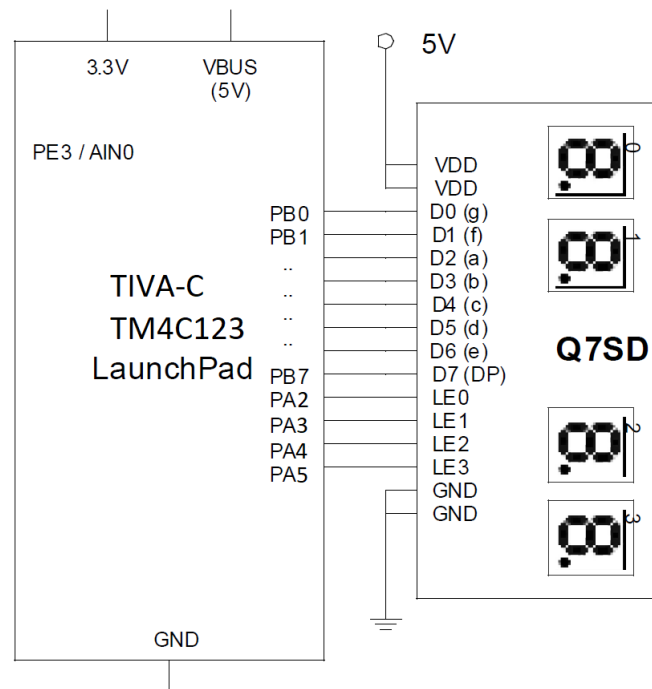


Figure 6.5: Connection to Seven Segment Display

Exercises

Program the LaunchPad and four-digit 7-segment display module to alternatively display “HELO” and after some delay, “2021”.